PS200C Causal Inference, Spring 2025

Introduction

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Modes of Statistical Inference

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   - means, correlations, regression coefficients
   - hypothesis testing
   - inferring “ideal points” from roll-call votes
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   - predicting future state failures from past failures and covariates
   - predicting vote share from a survey of likely-voters
3. Causal Inference: engages *counterfactuals*
   - inferring whether the lack of war among democracies can be attributed to regime type
   - determining whether a medical treatment improved outcomes.
   - generally, estimating effect of some policy or institution or treatment on some outcome...

You cannot escape

Actually it's more than that,

1. *Some* descriptive enterprises are truly descriptive, but many are at least hoping the reader thinks the result is "suggestive" or "consistent with" a causal implication.
2. Issues such as missingness, case-selection, choosing what to condition on, and generalization all actually invoke causal assumptions, even if your goals are descriptive.
3. If you can do a randomized experiment, great, but many questions (e.g. mediation) aren't answered by the randomization.

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In fact, I hope the course helps to convince you that technical sophistication of methods often does not solve core problems...its not about the latest estimator, machine learning, or more complicated standard errors.

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Stylized illustration:

- you have data on X and Y
- you want to learn some effect of X on Y
- you compute something with data, say, $coef(Y \sim X)$
- trust me (for now): *without further assumption*, any causal effect can produce any coefficient...and thus any coefficient you get is equally suggestive of any causal effect.

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Curious as to what it takes? What could be better than data? Take the class!

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- ... it is not the method or estimator, much less software, that makes the result causal.

Causal Identification Requires Assumptions

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Credibility does NOT derive from "what method you used" or how new it is

- ...It is about finding a technique whose assumptions you can work with
- There are not "causal methods"! You can't say "it's causal because I did xyz."
- This is one of the hardest things for many students to absorb, in part because we are surrounded by mistakes and false advertizing.

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- How can we understand how vulnerable a given estimate might be to threats such as confounding?
- **How can you conduct top quality work that seeks to make causal claims in tricky situations, while accurately characterizing what we can and cannot be claimed?**

This year's ambitious plan

Week	Topic	Assignment
1	Introduction; Potential Outcomes	Pset 0: Background Check
2	Potential Outcomes into Experiments	Pset 1: POs & ID in randomization
3	Conditional Ignorability (Identification)	Read+discuss example
4	Conditional Ignorability (Estimation)	Pset 2: CI in theory and practice; another paper
5	Directed Acyclic Graphs (DAGs)	Pset 3: DAGs; discuss another CI paper
6	Sensitivity Analysis for CI	Readings; Pset 4: Applying sensitivity analysis
7	(i) Difference-in-Differences and TWFE (ii) Synthetic control/ panel methods	Read 2 papers, discuss parallel trends
8	(i) Instrumental Variables (ii) Sharp Regression Discontinuity	Pset 5: Read and discuss 3 papers on assumptions; one estimation example
9	(i) Fuzzy Regression Discontinuity (ii) Overflow (or choose a topic)	—
10	Presentations (Part 1)	—
Final slot	Presentations (Part 2)	—

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A note on coding and why I don't emphasize it.

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  And we will use:
  - Hernan MA, Robins JM (2020). Causal Inference: *What If*. Boca Raton: Chapman & Hall/CRC."

  which you can fortunately get freely online:

  `https://cdn1.sph.harvard.edu/wp-content/uploads/sites/1268/2021/01/ciwhatif_hernanrobins_31jan21.pdf`.

Project

The final project pulls it all together!

- Work in small groups (2-3 usually, with some exceptions)
- Present to class in last week(s)
- **Classic project:** focus on a substantive question of interest to you and produce a cutting edge analysis that takes causal inference seriously, using some of the tools you learned in class.
- **Sensitivity study**. Using tools you will learn in class, apply sensitivity analysis to an important finding from your literature or otherwise of interest to determine "how much confounding would it take to alter this conclusion?"
- **Research practice round-up**. Review a discipline or literature to describe common practice, issues, successes, etc.
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Tentative milestones,

- April 26: submit teams and a notional topic for your project (we can help with team-finding as needed)
- May 10: submit a two-page progress report, schedule a meeting with us
- May 28-June 6 (depending on numbers): class presentations
- By June 14 (end of quarter) submit final project to website

Course Requirements and Grades

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- Psets (50%): about 5 of them,
  - Problem Set 1 is a review – will be up soon.
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- Project milestones: 10%
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Come to class, engage, do the work. If you aren't going to attend regularly, don't take the class.

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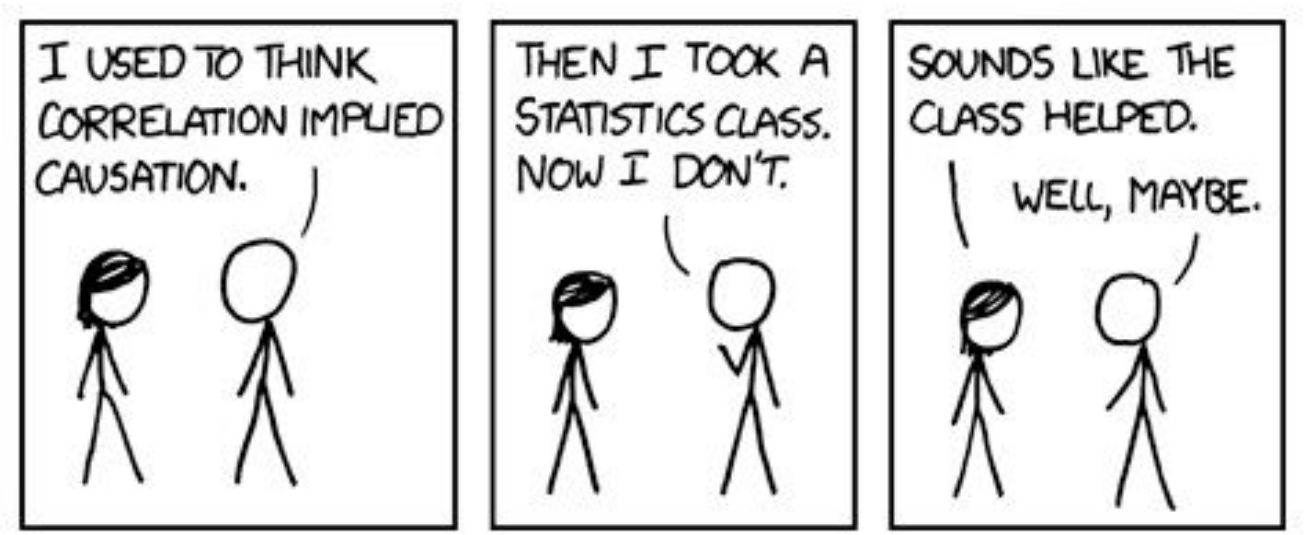
- Your TA: Barney

Some resources

1. CAPS: https://www.counseling.ucla.edu. They also have a 24/7 crisis support at 310-825-0768
2. The Bruin Fund offers funding for students with computing needs.
3. I'm always happy to help you get any services you need or to figure out whats available.



- Questions?
- Introductions!